Diseases of Jasmine

1. Rust - *Uromyces hobsoni*
2. Cercospora leaf spot/ blight - *Cercospora jasminicola*
3. Alternaria leaf spot - *Alternaria jasmini*
4. Anthracnose - *Elsinoe jasminae*
5. Colletotrichum leaf spot - *Colletotrichum jasminicola*
6. Phyllosticta leaf spot - *Phyllosticta jasminina*
7. Myrothecium leaf spot - *Myrothecium roridium*
8. Chlorotic ring spot - *Jasminum Chlorotic Ring spot Virus* (*JCRSV*)
9. Mosaic - *Jasmine Yellow Mosaic Virus (JaYMV*-AP)

Rust - *Uromyces hobsoni*

On under surface of leaf

b) On upper surface of leaf

Rust - *Uromyces hobsoni*

Rust - *Uromyces hobsoni*

The disease is wide spread in India and has been reported from Himachal Pradesh, Jammu and Kashmir, Maharashtra, Uttar Pradesh, Madhya Pradesh, Bihar, Karnatka and Rajasthan.

Symptoms:

- The fungus attacks **leaves, stem and inflorescences.**
- Leaves show the presence of orange coloured aecial cups on both sides, but **predominantly on the lower surfaces**
- The fructification or sori are formed **on the thickened portions**.

Rust - *Uromyces hobsoni*

Symptoms contd….

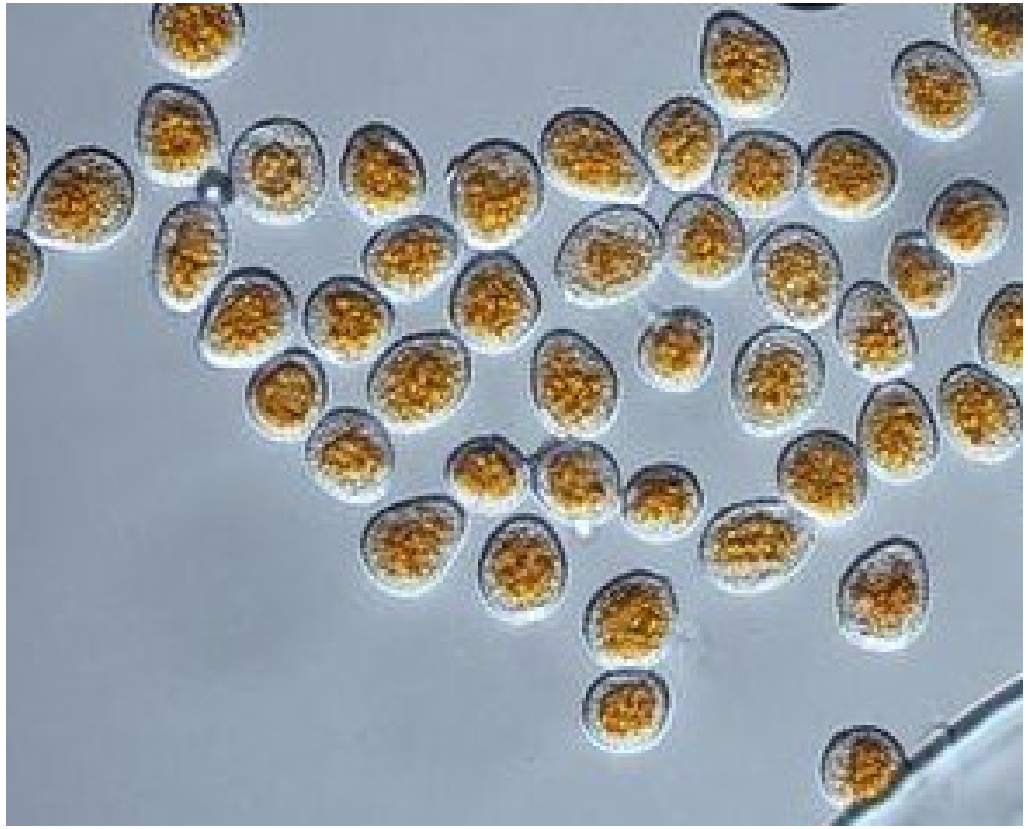
- The sori are initially **yellow owing to their contents, the aeciospores** later turn **darker due to formation of teliospores**.
- The infected portion becomes **hypertrophied, especially on the affected stems cankers, while on leaves blisters or swellings are formed**
- The affected portion later **thickened and gets puckered** and **flower buds do not open** in severe conditions.

Rust - *Uromyces hobsoni*

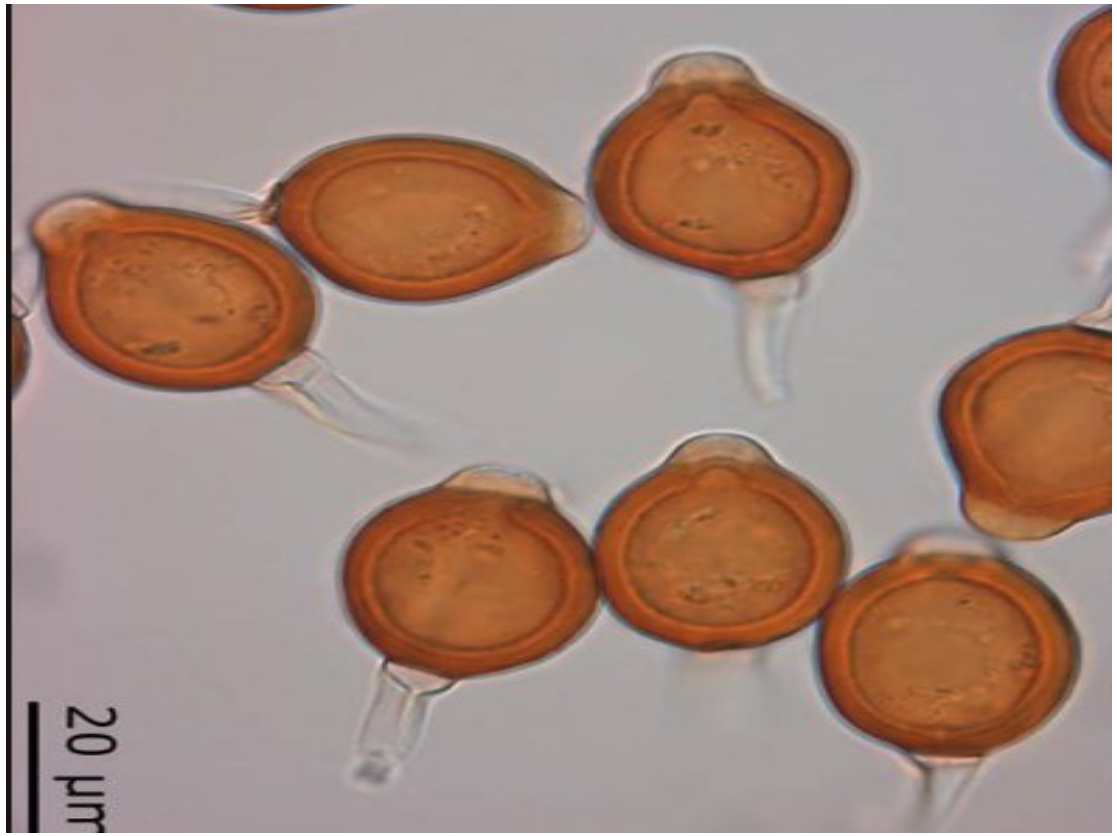
Pathogen:

- The pathogen produces aecia which are raised, scattered, orange-coloured, while **aeciospores** are orange-yellow, thin-walled, 16.5-21 x12-18 µm in size.
- The teleutospores are dark brown, one celled, mostly ovate, thick-walled, 35-45x20- 30 µm in size and pedicel is hyaline or yellow in colour.
- The **spermogonial stage is not observed.**

Rust - *Uromyces hobsoni*



Aeciospores are orange-yellow, thin-walled



The teleutospores are dark brown, one celled, mostly ovate, thick-walled, and pedicel is hyaline or yellow in colour.

Rust - *Uromyces hobsoni*

- Autoecious rust (Demicyclic rust/ Ospis type rust) in which Uredial stage is absent.
- **PSI**: Teliospores in infected plant
- **SSI:** Wind borne aeciospores
- **Demicyclic- Uredospores absent**

Rust - *Uromyces hobsoni*

Disease cycle and epidemiology

- The infection is more in winter and spring season.

Management:

- Collect and destroy the infected plant debris.
- With the initiation of the disease, spray the crop either with mancozeb (0.25%) or EBI fungicides like hexaconazole (0.1%) or difenoconazole (0.05%) and repeat at 10 days interval.

Cercospora leaf spot/ blight - *Cercospora jasminicola*

Cercospora jasminicola **infects** ***Jasminum officinale***

Cercospora leaf spot/ blight - *Cercospora jasminicola*

Fig. 3.2.1.2. Jasmine – Cercospora leaf spot

Cercospora leaf spot/ blight - *Cercospora jasminicola*

The disease is **widespread in its occurrence.**

In India, it has been reported from Punjab, Bangalore and Pune.

Symptoms:

- The symptoms appear as **circular to irregular, reddish- brown spots with a darker border on both surfaces of leaf** and on other aerial parts of the plant.
- Severe infection leads to premature defoliation which has an adverse effect on flowering.
- The pathogen also **attacks the stem and branches.**

Cercospora leaf spot/ blight - *Cercospora jasminicola*

Pathogen:

- *C. jasminicola* produces conidiophores which are amphigenous, arising in groups on a dark brown, thick-walled stroma, simple, septate, pale to olivaceous brown and geniculate with hyaline and narrow tips.

- However conidia are numerous, hyaline, multicelled with 9-13 transverse septa, 36-88 x 1.53-4.5µm, obclavate to cylindrical, straight to curved with obconically truncate base and subacute tip.

Cercospora leaf spot/ blight - *Cercospora jasminicola*

Disease cycle and epidemiology:

- The spread of the disease is **high during rainy season with slight high temperature and humidity regimes**.
- Disease cycle and epidemiology are explained in detail under **tomato and okra diseases**.

Management:

- Collect and burn the infected plant debris.
- With the initiation of the disease, spray the crop either with carbendazim (0.1%) or benomyl (0.1%) or mancozeb (0.25%), or copper oxychloride (0.3%) and repeat at fortnightly interval.

Alternaria leaf spot - *Alternaria jasmine, A. alternata*

***Alternaria jasmini* attacks**
J. grandiflorum, J. auriculatum

A. alternata

Alternaria leaf spot - *Alternaria jasmine, A. alternata*

Symptoms

Alternaria jasmini

- Symptoms on the upper surface of leaves appear as reddish brown circular spots (2-8 mm in dia.) concentric rings characteristics of which spread rapidly during rainy season.
- Oval to elongated brown spots develop on stem, petioles, sepals and petals
- In severe disease conditions, **vegetative buds and young branches dry up.**

Alternaria leaf spot - *Alternaria jasmine, A. alternata*

Symptoms

A. alternata

- Spots which are mostly marginal, start from the **tip of the leaves and proceeds towards petiole end.**
- Irregular wood brown coloured lesions are separated from healthy tissue by dark coloured bands.
- In lateral stage shot hole symptoms are visible

Alternaria leaf spot - *Alternaria jasmine, A. alternata*

Pathogen

- Conidia are muriform, beaked, brown, smooth and in chains.

Disease cycle and epidemiology:

- The spread of the disease is high during rainy season with slight high temperature and humidity regimes.
- Disease cycle and epidemiology are explained in detail under **tomato and okra diseases.**

Alternaria leaf spot - *Alternaria jasmine, A. alternata*

Management:

- Collect and burn the infected plant debris.
- With the initiation of the disease, spray the crop either with carbendazim (0.1%) or benomyl (0.1%) or mancozeb (0.25%), or copper oxychloride (0.3%) and repeat at fortnightly interval.

Anthracnose - *Elsinoe jasminae*

Anthracnose on shoots

Anthracnose - *Elsinoe jasminae*

Elsinoe jasminae

Anthracnose - *Elsinoe jasminae*

Symptoms

- Circular to angular, raised spots with slight depressions on the lower surface of the leaves which are pinkish buff in colour and 1-5mm in diameter.
- Similar spots appear on tender shoots also.
- Later these spots coalesce to form large patches.

Anthracnose - *Elsinoe jasminae*

Pathogen

- In *E. jasminae*, the ascomata is composed of pseudoparenchymatous epithecium and measured 45-110 x 20-45µm in size.

- Acervuli are intra-epidermal, erumpent, elongated to elliptical, light brown with the basal stroma giving rise to compactly grouped yellow conidiophores measuring 12-27 x 32-75µm in size.

Anthracnose - *Elsinoe jasminae*

Disease cycle and epidemiology

- The disease incitant overwinter as pseudothecium in infected plant debris.
- High humidity and moderate temperatures are favourable for the initiation and spread of the disease.

Management:

- Collect and destroy the infected plant debris.
- Spray the crop either with zineb (0.25%) or copper oxychloride (0.3%) and repeat at 10-14 days interval.

Colletotrichum leaf spot - *Colletotrichum jasminicola*

Colletotrichum leaf spot - *Colletotrichum jasminicola*

Symptoms

- **Large, circular, distinct and brownish to grayish spots** outlined with **brown to yellowish halo** appear on the upper surface of the leaf.
- The infected tissues turn grayish in due course, become papery and fall off.
- Spots are studded with numerous acervuli.

Colletotrichum leaf spot - *Colletotrichum jasminicola*

Pathogen

- In *Colletotrichum jasminicola*, acervuli are numerous, epiphyllous, erumpent, globose to cushion-shaped, 80-140 x 64-120 µm in size, setae are numerous, long, dark, thick and septate with pointed tips.
- Conidia are numerous, one celled, hyaline, oblong with rounded ends, guttulate, 8-15 x 4 - 4.5 µm in size.

Colletotrichum leaf spot - *Colletotrichum jasminicola*

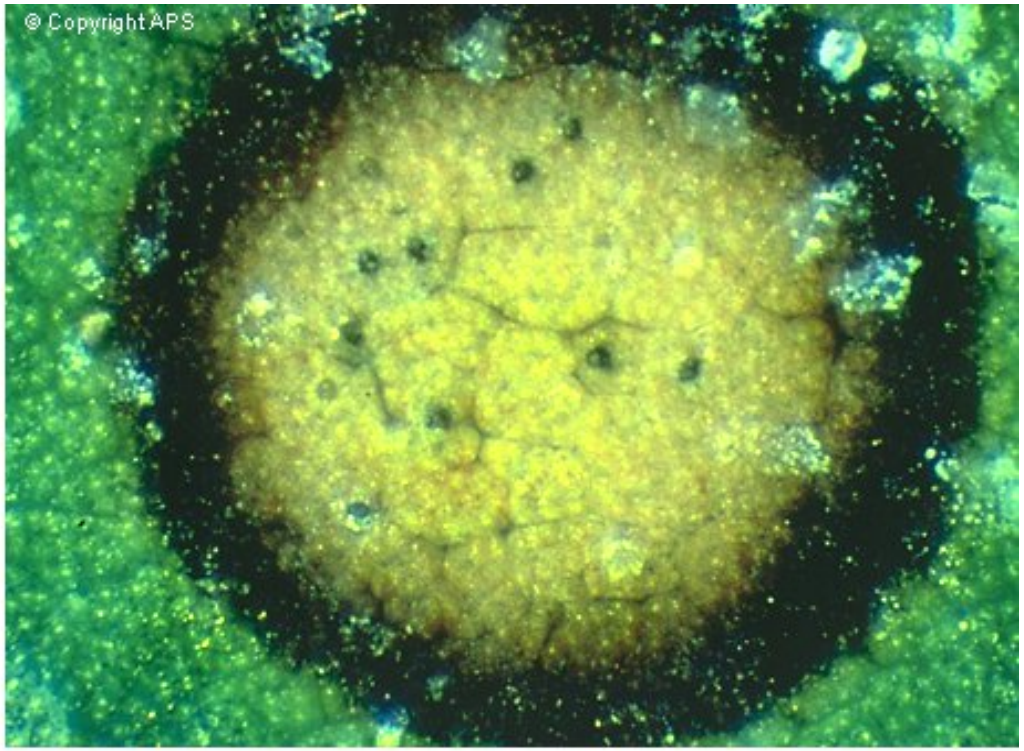
Disease cycle and epidemiology

- The disease incitants overwinter as perithecium in infected plant debris.
- High humidity and moderate temperatures are favourable for the initiation and spread of the disease.

Management:

- Collect and destroy the infected plant debris.
- Spray the crop either with zineb (0.25%) or copper oxychloride (0.3%) and repeat at 10-14 days interval.

Phyllosticta leaf spot - *Phyllosticta jasminina*



Phyllosticta leaf spot with pycnidia.

Phyllosticta leaf spot - *Phyllosticta jasminina*

Symptoms:

- Infectious spots are usually **circular, pale, fulvous to brownish** with smooth margins.
- Adjacent spots become large by coalescing together.

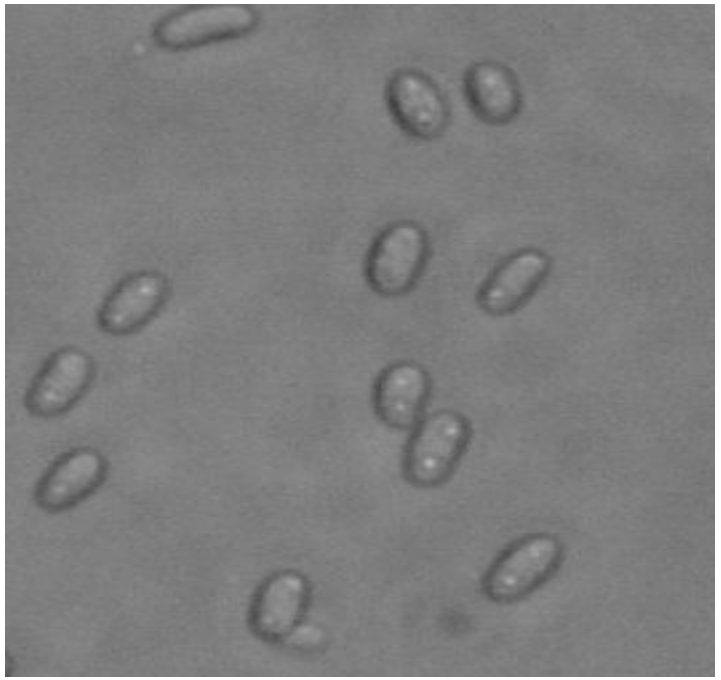
Phyllosticta leaf spot - *Phyllosticta jasminina*

Pathogen

Phyllosticta jasminina

- Pycnidia are epiphyllous, ampulliform to sub-globse, later erumpent, ostiole 90-170 µm in diameter

- Spores are hyaline, single- celled, ovoid to oblong 5.5-8.5 x 3.3-4.5 µm in size.



Phyllosticta leaf spot - *Phyllosticta jasminina*

Disease cycle and epidemiology

- These pathogens overwinter as **perithecium** in infected plant debris.
- The disease spread is favoured by temperatures ranges between 20 - 25°C along with high humidity.

Management:

- Collect and destroy the infected plant debris.
- Avoid dense planting, shading of plants, and excessive watering of the beds and fields.
- Spray the crop with carbendazim (0.1%) or mancozeb (0.25%) and repeat at 14 days interval.

Myrothecium leaf spot - *Myrothecium roridium*

Ring spot

Myrothecium leaf spot - *Myrothecium roridium*

Symptoms:

On leaves, symptoms appear as **circular to irregular, brown to dark brown spots** surrounded by yellowish halo **but no sporulation is observed on these spots.**

Pathogen

- Conidia are aseptate, pigmented, black in mass, wet when young, drying to become hard, shiny black, convex with a white halo of mycelium, 7.2 x 1.8 µm in size.

Myrothecium leaf spot - *Myrothecium roridium*

Disease cycle and epidemiology

- The pathogen survives in infected plant debris
- Moderate temperature and high relative humidity (wet weather) conditions favours the disease.

Management

- Spraying the crop with carbendazim (0.1%) or zineb (0.25%) at 10-14 days interval is advocated for the control of this disease.

Chlorotic ring spot - *Jasminum Chlorotic Ring spot Virus* (*JCRSV*)

Mosaic - *Jasmine Mosaic Virus (JMV)*

Symptoms

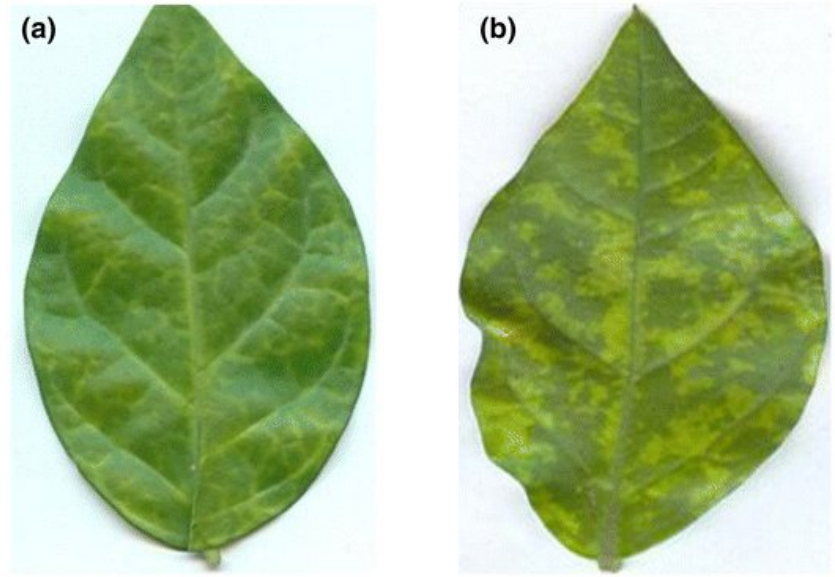
- The diseased plants show stunted growth and yellowish green appearance with small leaves.
- Yellowish **green to chlorotic flecks of 1-2 mm diameter** appear irregularly on the leaf.
- The streaks later form ring like structure which are more conspicuous on the older leaves.
- The diseased plants show stunting and yellowish green appearance with small leaves.

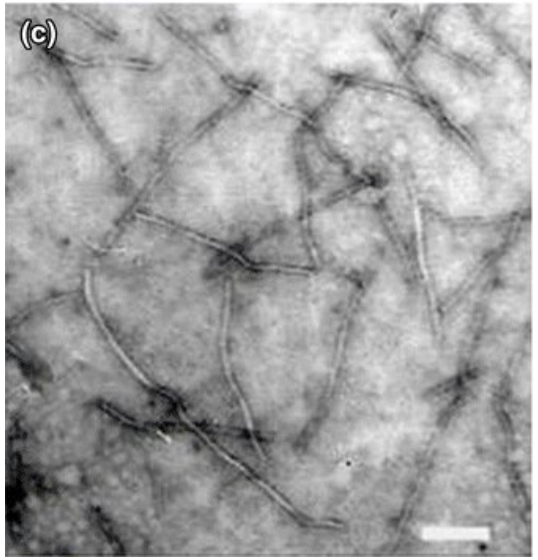
Chlorotic ring spot - *Jasminum Chlorotic Ring spot Virus* (*JCRSV*)

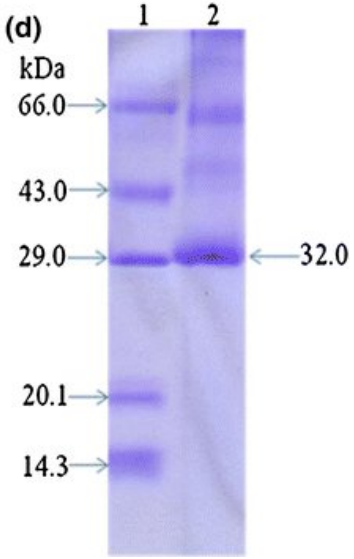
Symptoms

Whitefly (*Bemisia tabaci*) transmitted chlorotic ringspot virus on jasmine observed in different parts of Kerala.

Mosaic - *Jasmine Yellow Mosaic Virus (JaYMV*-AP)







Mosaic - *Jasmine Yellow Mosaic Virus (JaYMV*-AP)

Symptoms

- Vein clearing
- Spots intermingled with normal green colour with yellow rings with mosaic pattern.

Mosaic - *Jasmine Yellow Mosaic Virus (JaYMV-*AP)

Pathogen:

- *Jasmine Mosaic virus - Potyviridae*

Transmission:

The virus is **graft transmitted** on *Jasminium simplicifolium, J. sambac* and *J. pubescens.*

Management:

- Destruction of the infected material and use of healthy stock
- ***Jasminum sambac* and *Jasminum pubescens*** are found to be **highly susceptible to the disease**.
- ***J. auriculatum*** is found **relatively tolerant** to the mosaic virus.